get_status

Command: `get_status`

Description

Retrieves **real-time health and status** of the scanner — battery, temperature, radio, power, clock sync, and terminal connection.

Command details:

- Pattern Name: Reader Health and Status Retrieval
- Communication Type: Bidirectional (Cloud to Device, Device to Cloud)
- MQTT Version: 3.1.1
- API Version: V1.1
- Document Version: 1.0.0
- Last Updated: 2026-03-12
- Applies To: RFD40 Series, RFD90 Series

Response fields:

Field	What it tells you
`powerSource`	How the scanner is powered — DC, Wall Charger, USB, or Cradle
`radioActivity`	Whether the RFID radio is currently active or idle
`radioConnection`	Whether the radio module is connected or disconnected
`hostname`	The scanner's network hostname
`systemTime`	Current device clock in ISO 8601 format
`temperature`	Internal temperature in °C — monitor for overheating
`ntp`	Clock sync status — offset (ms drift) and reach (sync success history)
`terminalConnection`	Paired phone/tablet — connection status and type (Bluetooth, USB, CIO)
`batteryStatus`	Battery capacity (mAh), charge %, health (Good/Average/Poor), charge state

Typical use cases:

- Dashboard monitoring — show live scanner health on a management console.
- Battery management — track charge levels across a fleet before a shift starts.
- Troubleshooting — check if radio is connected, temperature is normal, clock is synced.

MQTT topics for this command:

- Command topic (Cloud to Device):

`<tenantId>/CTRL/clients/cmnd/<deviceSerial>`

- Response topic (Device to Cloud):

`<tenantId>/CTRL/clients/resp/<deviceSerial>`

Example:

- Command: `zebra/CTRL/clients/cmnd/RFD40-12345678`
- Response: `zebra/CTRL/clients/resp/RFD40-12345678`

Ensure these topic paths match your configured endpoint topics.

Request

Request Fields

Field	Type	Required	Description	Example
`command`	string (enum)	Yes	Must be `get_status`. Tells the scanner to return its current health and status.	`get_status`
`requestId`	string	Yes	A unique ID you assign to this request. The scanner echoes it back in the response so you can match requests to responses.	`abcd123`

Request Examples

Generated Example

```
{
  "command": "get_status",
  "requestId": "abcd123"
}
```

Response

Response Fields

Field	Type	Required	Description	Example
`command`	string	Yes	Echoes back the command name — always `get_status` for this response.	`get_status`
`requestId`	string	Yes	Echoes back the requestId you sent, so you can match this response to your original request.	`abc123`
`apiVersion`	string (enum)	Yes	The API version of the IoT Connector responding to this command.	`V1.1`
`deviceStatus`	object	No	Real-time device health — power source, RFID radio state, temperature, battery level and health, NTP clock sync, and paired terminal (phone/tablet) connection.	
`deviceStatus.powerSource`	string (enum)	Yes	The source of power for the device (e.g., DC, Wall Charger, USB, or Cradle).	`CRADLE`

Field	Type	Required	Description	Example
`deviceStatus.radioActivity`	string (enum)	Yes	The activity status of the device's radio (e.g., Active or Inactive).	`ACTIVE`
`deviceStatus.radioConnection`	string (enum)	Yes	The connection status of the device's radio (e.g., Connected or Disconnected).	`CONNECTED`
`deviceStatus.hostname`	string	Yes	The hostname of the device on the network.	`RFD40-212735201D0053.local`
`deviceStatus.systemTime`	string	Yes	The system time of the device in ISO 8601 format.	`2022-07-01T03:01:31.464Z`
`deviceStatus.temperature`	integer	Yes	The current temperature of the device in degrees Celsius.	`32`
`deviceStatus.ntp`	object	No	Details about the Network Time Protocol (NTP) synchronization.	
`deviceStatus.ntp.offset`	integer	No	The offset in milliseconds between the device's clock and the NTP server.	`120`
`deviceStatus.ntp.reach`	integer	No	The reachability score of the NTP server.	`321`
`deviceStatus.terminalConnection`	object	No	Details about the terminal connection to the device.	
`deviceStatus.terminalConnection.status`	string (enum)	No	The connection status of the terminal (e.g., Connected or Disconnected).	`connected`

Field	Type	Required	Description	Example
`deviceStatus.terminalConnection.type`	string (enum)	No	The type of terminal connection (e.g., Bluetooth, CIO, or USB).	
`deviceStatus.batteryStatus`	object	No	Information about the battery status of the device.	
`deviceStatus.batteryStatus.capacity`	integer	No	The battery capacity remaining in mAh.	`21`
`deviceStatus.batteryStatus.stateOfHealth`	string (enum)	No	The overall health of the battery (e.g., Good, Average, or Poor).	`GOOD`
`deviceStatus.batteryStatus.chargePercentage`	integer	No	The percentage of battery charge remaining.	`23`
`deviceStatus.batteryStatus.chargeStatus`	integer	No	The charging status of the device battery: 0 - Charger not connected, 1 - Charger connected and charging in progress, 2 - Charger connected and battery is 100%.	`1`
`response`	object	Yes	Status of the command execution — check `code` to confirm success (0) or identify the error.	

Field	Type	Required	Description	Example
`response.code`	integer	Yes	Response code indicating success or failure. • **0** — Success • **1** — Command payload accepted • **2** — Invalid payload • **3** — Not able to retrieve information • **4** — Firmware update in progress • **5** — Can't reboot, inventory in progress • **6** — Region is not configured • **7** — Interface is not available • **8** — Insufficient flash size • **9** — File not found • **10** — Configuration already exists • **11** — Inventory in progress • **12** — No radio operation in progress • **13** — Firmware update failed • **14** — Battery too low for firmware update • **15** — WIFI: SSID not found • **16** — WIFI: Cannot delete	`0`

Field	Type	Required	Description	Example
			active SSID • **17** — WIFI: SSID missing • **18** — WIFI: SSID already exists • **19** — WIFI: SSID count overflow • **20** — WiFi is not supported • **21** — Certificate not found • **22** — Advanced configuration not set • **23** — Invalid enum value • **24** — Max 32 prefilters limit exceeded • **25** — Max 3 publish topics exceeded • **26** — Max 1 subscribe topic exceeded • **27** — Invalid tenant ID length • **28** — Tag match pattern length exceeded	
`response.description`	string	Yes	Human-readable description of the response code (e.g. "Success", "Invalid payload").	`Success`

Response Examples

Device status info

```
{
  "command": "get_status",
  "requestId": "abc123",
  "apiVersion": "V1.1",
```

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```
  "deviceStatus": {
    "powerSource": "USB",
    "radioActivity": "INACTIVE",
    "radioConnection": "CONNECTED",
    "systemTime": "2024-02-26T13:45:53.728Z",
    "temperature": 32,
    "ntp": {
      "offset": 0,
      "reach": 0
    },
    "terminalConnection": {
      "status": "CONNECTED",
      "type": "USB"
    },
    "batteryStatus": {
      "capacity": 6400,
      "stateOfHealth": "GOOD",
      "chargePercentage": 100,
      "chargeStatus": 1
    }
  },
  "response": {
    "code": 0,
    "description": "Success"
  }
}
```